logarithmic anymore (like it was for *BST*s). To perform a more rigorous analysis, let's again denote as `Tk(n)` the running time for this method, where `k` is the dimensionality of the points' space, `n` is the number of nodes in the tree; when we look more closely at each conditional fork, if we assume that we are working on a balanced tree, then:

- Each of the conditional at lines #13 and #14 would trigger a recursive call on approximately half the nodes, and so require time `Tk(n/2)`.
- If we enter the code blocks after conditionals at lines #2 or #12, those only require constant time.
- Conditionals at lines #4 and #8 both will run code blocks that requires creating a new node, `O(1)`, running `findMin`, `O(n1-1/k)`, and recursively call `remove`.